

Real-Time ECG Monitoring System

Project report submitted

In partial fulfillment of the requirements

For the award of the Degree of

Bachelor of Technology In Electronics and Telecommunication Engineering

by

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2022BEC153

Under the guidance of

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Declaration

I Rutuj Kurundkar [2022BEC153] hereby declare that this project work titled “**Real-Time ECG Monitoring System**” is carried out by me in the Department of Electronics and Telecommunication Engineering of Shri Guru Gobind Singhji Institute of Engineering & Technology, Vishnupuri, Nanded. The work is original and has not been submitted earlier whole or in part for the award of any degree/diploma at this or any other Institution / University.

Mr. Rutuj Kurundkar
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Date: / / 2026
Place: Nanded

Certificate

This is to certify that the project titled “**Real-Time ECG Monitoring System**”, submitted by **Rutuj Kurundkar** in partial fulfillment of the requirements for the 8th Semester Major Project of **Bachelor of Technology in Electronics and Telecommunication Engineering**, SGGSIE&T, Nanded. The work is comprehensive, complete and fit for final evaluation.

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Abstract

The rapid escalation of cardiovascular diseases necessitates a transition from periodic clinical check-ups to continuous, proactive heart health surveillance. This project presents the design and implementation of a **Real-Time ECG Monitoring System**, a wearable biomedical solution engineered to detect cardiac abnormalities as they occur.

The system aims to bridge the gap between home-based care and professional medical diagnosis by providing a reliable platform for long-term data acquisition.

The system architecture integrates a high-precision analog front-end with an embedded microcontroller to capture, filter, and process bioelectrical signals from the heart. By leveraging **Internet of Things (IoT)** protocols and **Cloud Computing**, the device facilitates the seamless transmission of live ECG waveforms to a secure web-based dashboard. This integration allows for real-time data visualization, ensuring that vital cardiac information is accessible to healthcare providers regardless of the patient's location.

The methodology encompasses the development of digital noise-reduction filters for signal clarity and the deployment of a real-time R-peak detection algorithm for accurate heart rate calculation. A core component of the software involves an automated alert system designed to identify and flag critical events, such as tachycardia or bradycardia, immediately. These features ensure that the system does not just record data but actively monitors for life-threatening irregularities.

Performance evaluation conducted under various physical conditions confirms the system's reliability in maintaining signal integrity and transmission stability. The results indicate that the developed prototype provides medical-grade data and immediate responsiveness, meeting all functional requirements for remote patient monitoring. Ultimately, this work contributes to **Biomedical Engineering** by offering a scalable, cost-effective solution for large-scale healthcare applications.

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ABBREVIATIONS

ECG - Electrocardiography

IoT - Internet of Things

R-peak - Point where the peak of the QRS complex occurs on the ECG waveform

mHealth - Mobile Health

HRV - Heart Rate Variability

AFE - Analog Front End

AD8232 - ECG signal conditioning integrated circuit

Ag/AgCl - Silver/Silver Chloride

SNR - Signal-to-Noise Ratio

RLD - Right Leg Drive

DSP - Digital Signal Processing

SA Node - Sinoatrial Node

AV Node - Atrioventricular Node

PR Interval - Time interval between atrial activation and ventricular activation

QRS Complex - Depolarization pattern in the ventricles

ST Segment - Time interval between ventricular depolarization and repolarization

Hz - Hertz

ADC - Analog to Digital Converter

MCU - Microcontroller Unit

I/O - Input/Output

RT ECG MS - Real-Time ECG Monitoring System

TCP/IP – Transmission Control Protocol / Internet Protocol

HTTP – Hypertext Transfer Protocol

MQTT – Message Queuing Telemetry Transport

TCP – Transmission Control Protocol

UDP – User Datagram Protocol

USB – Universal Serial Bus

1 INTRODUCTION

1.1 Background and Motivation

The present-day scenario in global healthcare shows that a digital revolution is taking place in this field. There is a need for the transformation from the current approach of reactive to proactive medical treatment for patients. Diseases of the cardiovascular system are among the primary causes of death in the world due to their hidden character, which usually reveals itself only during a serious event. Classic diagnostic methods based on regular hospital check-ups for resting electrocardiograms are unable to provide reliable results concerning transient anomalies occurring during normal activity. In the context of the development of distributed healthcare solutions, there is a growing requirement for continuous monitoring technologies that operate on a round-the-clock basis. The use of IoT and Embedded Systems can provide for the precise analysis of data related to patients' hearts.

Because of the unpredictable nature of cardiovascular conditions, integration in real time is no longer optional – it is a necessity. While traditional Holter monitors may be portable, they need several days of data collection prior to the analysis of those results by a physician, which can pose significant risks due to the resulting "information gap." In such circumstances, an integrated Real-Time ECG Monitoring System becomes immediately relevant for facilitating communication between the patient and the healthcare professional. Utilizing high sensitivity biosensors and wireless communications allows automatic detection of any disturbances in the heartbeat rhythm, such as arrhythmia or tachycardia.

In addition to the detection capabilities, the use of such technology is based on its potential to help reduce the strain on the healthcare facilities. Readmissions of the patients into hospitals for cardiac monitoring is a resource-consuming process. The implementation of such monitoring helps solve this problem by detecting initial symptoms of heart failure or risks of having a stroke before the patient needs immediate assistance. Such solutions improve the efficiency of the patient-doctor collaboration so that only acute patients will be admitted to the hospital whereas chronic ones will receive the necessary treatment at home. Finally, the move towards mHealth solutions empowers patients to monitor their conditions independently.

The present-day generation of ECG monitoring is revolutionary from a technical point of view in terms of signal processing and analysis. While previous generations of analog-based devices were susceptible to interference, the latest generation of digital monitors effectively filters out environmental noise while consuming minimal amounts of energy. Modern ECG monitoring systems can analyze complicated physiological signals, compare them to baseline readings, and securely send their health records to the cloud. The recent breakthroughs in Machine Learning have contributed to this by enabling devices to detect anomalies even if they fall outside the purview of standard algorithms.

1.2 Objective of the Project

The main objective of this study is to create and implement an ECG monitoring system which can measure the activity of the heart and display the ECG signal continuously. This system should be able to acquire and analyze the signal in such a way that it is accurate and clear.

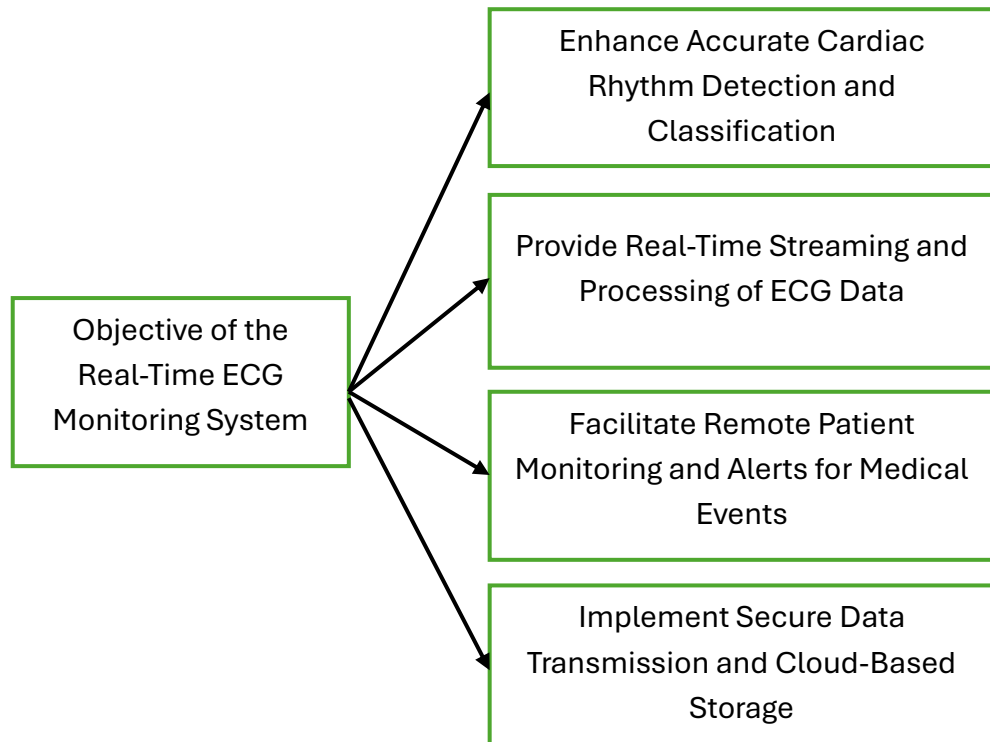


Figure 1.1: Objective of the Project

1.2.1 Continuous Acquisition of ECG Signal

This is to guarantee that the electrical impulses coming from the heart of humans are being continually acquired using the ECG sensors that are connected to the signal conditioning circuitry. The ECG signals are continuously acquired in real-time while monitoring the patient. Continuous acquisition enables the continual observation of the activities of the heart without any disruption.

1.2.2 Perform Real-Time ECG Signal Processing

This is aimed at processing the ECG signal captured by using both hardware and software methods. This involves filtering out any interference present in the ECG signal, thereby making the ECG signal more accurate and clear. The processed ECG signal is then converted to digital form.

1.2.3 Display ECG Waveform and Heart Rate Information

This is aimed at showing the ECG waveform together with the calculated heart rate on a display monitor like an LCD display unit or computer screen. Through this system, there is a visualization of both the waveforms as well as the heart rates of patients. This will allow health professionals or end users to view the status of the heart of patients.

1.2.4 Ensure Reliable and Stable System Operation

The aim is to ensure consistent performance of the system during constant monitoring. The system runs without any disruption while measuring and

presenting the ECG data. Consistent performance makes it possible for the heart activity to be monitored accurately.

1.2.5 Support Basic Cardiac Monitoring in Healthcare Environments

The aim is to develop an operational monitoring system that can be implemented in hospitals, clinics, laboratories, or even in-home healthcare facilities. This technology facilitates periodic monitoring of heart signals and assists healthcare professionals in monitoring patients' heart functions.

1.3 Scope of the Project

The scope of this project includes the creation and implementation of the entire system of Real-Time ECG Monitoring System aimed at providing constant monitoring of the heart condition. It implies not only designing the hardware and software structure but also establishing the operating conditions and diagnostic purposes of the proposed system. Among the aspects of the scope are:

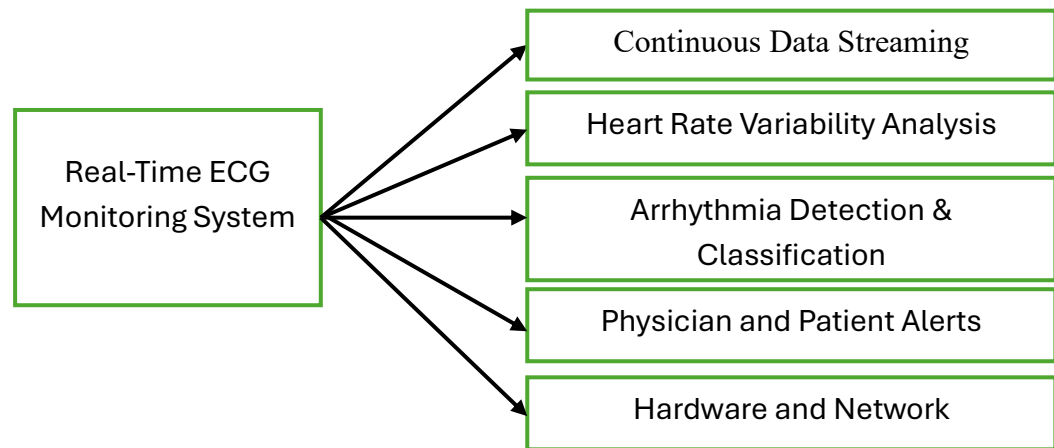


Figure 1.2: Scope of the Project

1.3.1 System Capabilities

The ECG system has been developed as a modular design which is intended to integrate wearable hardware with digital health systems. This involves developing wireless communication channels for the purposes of interacting with the various interfaces, such as the application interfaces, the cloud database interfaces, and the interface used by physicians. In order to facilitate this process, there is an emphasis on ensuring the compatibility of the system since this involves developing the system in such a way that it can operate in the everyday environment of the user.

1.3.2 Focus on Cardiac Monitoring

One of the key areas within the scope of this system is the improvement of the detection of transitory cardiac events. The system is designed to collect different types of information about the ECG, such as heart rate variations, intervals between R-peaks, and heartbeat patterns. In this context, the system examines real-time data to detect early signs of heart diseases like arrhythmia or tachycardia. For instance, identifying any changes that

occur in relation to normal values, the system can motivate people to consult with a healthcare specialist or inform their emergency contacts.

1.3.3 Data Visualization and Patient Analytics

The project scope includes the development of diagnostic instruments for telemedicine. The software analyzes the raw electric signal data and presents the information in a visual format like waveform data and heart rate information. Medical practitioners are given an overall record for planning purposes. The scope involves designing algorithms that automatically analyze the signal data and generate brief reports or warnings. This feature allows medical practitioners to respond more quickly to changes in the patient's health state.

1.3.4 Signal Integrity and Performance Optimization

An additional element of the scope of this project is improving the quality of signals and power efficiency for wearables. The hardware contains filters to regulate noise, including muscle tremor or power-line interference. The scope of this project involves developing power-efficient designs that prolong battery life but still achieve high sampling speeds. Using simulations of different physiological states, the system is programmed to ensure optimal signal stability in motion. The expected result is minimizing false alarms and maximizing the reliability of heart activity data.

1.3.5 Technical and Regulatory Boundaries

This scope of work involves ensuring data privacy and security in accordance with medical regulations. Since the physiological data is very sensitive, the security features like end-to-end encryption and user authentication have been used in the system to make sure that when the cardiac records are transmitted by the system, the patient's confidentiality is maintained. The security aspects of the system will adhere to certain guidelines so that it maintains its reputation as an accurate diagnostic tool. The scope also entails logging within the cloud platform, ensuring that health alerts are accounted for.

1.4 Structure of the Report

This report is subdivided into eight parts, each describing a particular facet of the Real-Time ECG Monitoring System. The structure of the paper is logical and follows a particular order—from fundamental aspects to hardware description and testing.

The first chapter, Introduction, provides the rationale for this project and describes its goals and scope. In this part, you will find information about the problem solved by the system—the necessity for continuous cardiac monitoring.

The second chapter, Literature Review, focuses on modern developments in wearable medical devices. You will learn about different types of ECG monitoring systems, communication channels used for data transfer, as well as automated algorithms for arrhythmia detection. The limitations of modern Holter monitors are discussed.

In chapter three, Theoretical Concepts, you will find theoretical aspects of the project development. The topics include basic concepts of heart electrophysiology, signal processing theory related to noise elimination, as well as mathematical concepts necessary for the detection of R-peak and HRV calculation.

Chapter four, Methodology, describes the approach followed during the development of the project and includes the design of the hardware and software, as well as the choice of sensors and microcontrollers.

The fifth chapter, Implementation and Execution, details the physical construction of the prototype and installation of software code.

The sixth chapter, Experimental Results and Comparison, outlines the performance metrics of the device, which include the signal-to-noise ratio and accuracy of the automated heartbeat alert systems. The comparison is done using the outputs from the prototype and clinically acceptable electrocardiography outputs.

The seventh chapter, Discussion and Analysis, evaluates the results, emphasizing the strength of the real-time nature of the device and its shortcomings in terms of battery endurance or interference when the patient is involved in intense activities.

Lastly, chapter eight, Conclusion and Future Scope, reviews the results and proposes potential areas for improvement, such as the inclusion of machine learning to automate the diagnostics process and create a more compact device that can be worn.

2 LITERATURE REVIEW

2.1 Introduction to Real-Time ECG Monitoring Systems

The domain of cardiac medicine has witnessed an evolving trend toward the use of digital technologies and wearable devices, focusing on transitioning from occasional clinical examinations to continuous monitoring. Current research demonstrates that the utilization of remote monitoring devices is revolutionizing early identification of heart ailments and clinical decision-making concerning patients' health [8][9]. Within the biomedical engineering realm, embedded systems and signal processing technologies are used to analyze biological electrical signals and identify abnormalities of heart rhythms, which are not detected using conventional approaches. Similar findings were observed in a review study, indicating that real-time electrocardiograms assist both urgent medical treatment and daily monitoring of heart disease in a distributed setting [16]. This chapter provides an overview of relevant literature concerning real-time monitoring in cardiac medicine, particularly regarding (a) the hardware sensor technology, including electrode materials and analog front-end design, and (b) software environments such as wireless communication, cloud computing, and automatic detection of irregularities.

2.2 Evolution of Wearable Cardiac Technology

Traditionally, heart monitoring was carried out using large medical devices or even the holter device which had drawbacks in terms of limited recording periods and latency in data processing. Real-time heart monitoring solves these shortcomings through offering real-time visualization and reaction. In recent years, the development of low-energy biosensors has made it possible to have a wearable device without losing accuracy in data transmission.

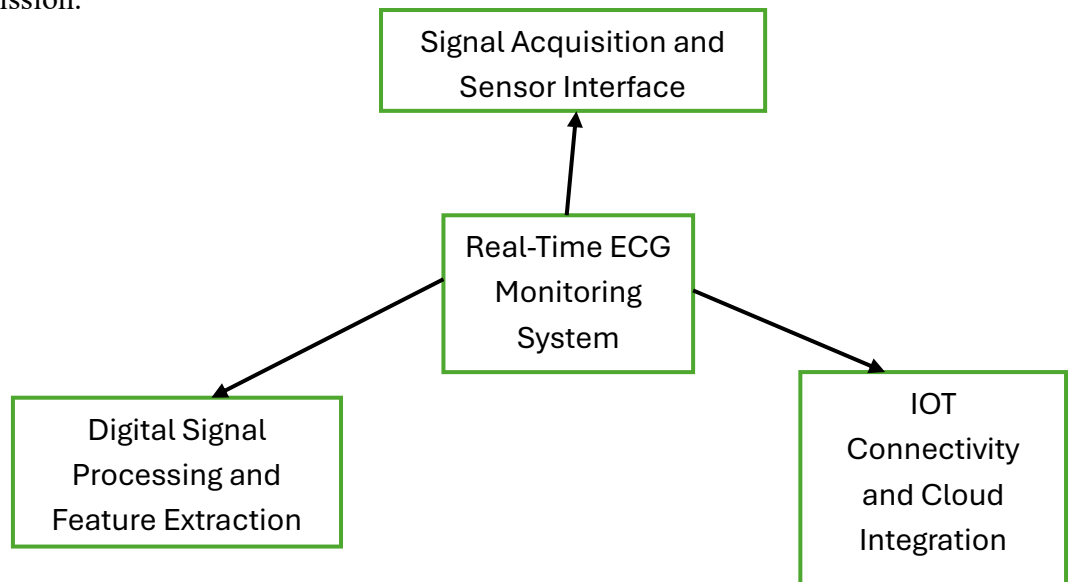


Figure 2.1: Real-Time ECG Monitoring System Architecture

2.2.1 Acquisitions of Signals and Sensors

The first and foremost step in an ECG acquisition process is the correct capturing of the myocardium's biopotentials. Advanced sensors and Analog Front Ends (AFE) like the AD8232 are employed to amplify low-level signals while simultaneously suppressing common mode noise. It has been established from literature that the choice of electrode whether Ag/AgCl type or dry capacitive plays a major role in SNR. It has been observed by many biomedical engineers that pre-processing the signals with the help of instrumentation amplifier and RLD circuits is crucial in order to avoid interference caused by power line and baseline wandering.

2.2.2 Digital Signal Processing and Feature Extraction

After digitization, the DSP algorithms are used to recognize important cardiac markers. The Pan-Tompkins algorithm is one of the most frequently referenced techniques for detecting R-peaks, which then becomes the starting point for estimating the heart rate and HRV. Software filters are used to eliminate any remaining artifacts in the signal, including muscle tremor noise. Research indicates that successful feature extraction leads to accurate classification of cardiac rhythms into different classes, such as normal sinus rhythm, tachycardia, or bradycardia. Through automation, the system eliminates the need for manually analyzing the strips, offering instant results regarding the patient's cardiac health.

2.2.3 Cloud Integration and Remote Monitoring

The DSP algorithms are utilized for identifying key cardiac markers following digitization. One of the most commonly cited techniques is the Pan-Tompkins algorithm, which involves detection of the R-peaks that are used as a basis for determining the heart rate and HRV. Filters embedded in the software are employed for removal of artifacts present in the signal, including muscle tremor. It has been found through research that proper extraction of features results in correct classification of the cardiac rhythm into various categories, such as normal sinus rhythm, tachycardia, or bradycardia. The automation process ensures that there is no need to analyze the strips manually anymore.

In summary, the literature proves that combining hardware procurement with cloud analytics fosters a proactive diagnosis system. Although hardware guarantees accurate signals, it is the software and cloud aspects that make scalability possible for contemporary medicine.

2.3 Remote Patient Monitoring and Diagnostics

Cardiac monitoring is an integral but information-heavy component of modern cardiac management. Traditional diagnostic tests are based on short-term data that constitute only a fraction of the day's activities and, thus, overlook cardiac events. Technology is being used to enhance various facets of cardiac monitoring and emergency response, seeking to provide quality information while ensuring that patients remain

mobile. Recent studies have reported the use of technology in continuous data acquisition, automatic rhythm recognition, and emergencies notification.

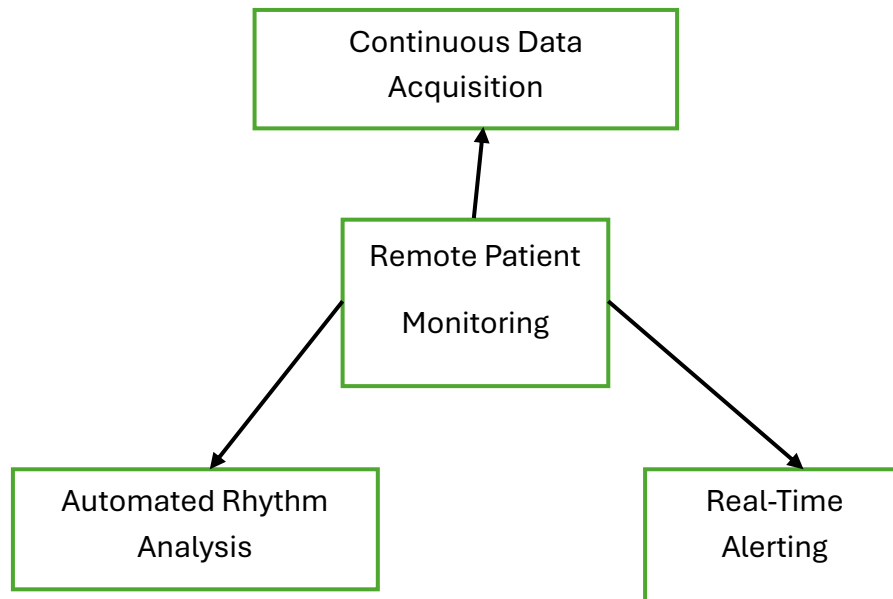


Figure 2.2: Elements of the ECG Monitoring Ecosystem

2.3.1 Continuous Data Acquisition and Streaming

A critical challenge in cardiology is the detection of sporadic rhythms. This process becomes easier with digital means through the use of wearable sensors to measure the electrical activity of the patient while performing their regular activities. Studies show that Internet-of-Things technology is applied in streaming raw ECG signals to cloud databases. Embedded systems can easily manage the sampling process, which demands very high speeds compared to intermittent tests done manually. The continuity of data provided through these systems enables health professionals to make diagnoses faster.

2.3.2 Automated Rhythm Analysis and Classification

Moreover, digital algorithms have played an important role in intelligent cardiology diagnosis as well. Predictive modeling will allow for better detection of certain waveform patterns, such as detection of the P-wave or QRS complex for determination of heart rate variability (HRV). The system will have an ability to recognize the patterns based on various physiological circumstances; thus, the algorithm will be able to distinguish between normal sinus rhythm and any other irregularities. In addition, predictive modeling will benefit greatly from signal processing techniques, whereby certain predictions can be made mid-way through the observation process.

2.3.3 Real-Time Alerting and Clinical Intervention

Through monitoring, technology ensures patient safety due to the availability of instant responses to any safety concerns. For instance, algorithms can analyze ECG data

streams almost instantly to detect any anomaly, such as increased or decreased heartbeats. Moreover, technology enables better use of the remote monitoring process by detecting problems with electrodes or battery charge level on the device. The advantage of using technology is that, in a decentralized health care system where patients are monitored at their homes, out-of-range values could be detected easily and reported immediately.

3 THEORETICAL CONCEPTS

3.1 Introduction to Electrocardiography

The electrocardiography process involves the biomedical measurement of the electrical signals emitted by the heart during physiological activities in humans. This is because the heart acts as one of the essential organs in the body, which ensures that the circulation of blood takes place efficiently. The heart works in coordination with an electrical system, which generates electrical impulses within it and stimulates each beat. Such impulses are created by certain specialized cardiac cells and pass through conduction pathways.

The electrical energy generated by the heart through each cardiac cycle can be detected on the surface of the skin since the human body conducts electricity. The signals are known to have small amplitudes and particular frequency components. The strength of the ECG signal ranges from a few millivolts; hence, the detection of such signals demands high sensitivity and accuracy. Conductive material electrodes are positioned on the body at predetermined locations to detect the signals.

These electrical signals, after detection, are fed into a signal conditioner circuit. Here, the signals are amplified and filtered to enhance the quality of the signals and reduce unwanted noise. These signals are then converted to digital format via an Analog to Digital Converter. These digital signals are displayed in the form of a waveform that depicts the electrical events taking place in the heart. This waveform is referred to as an Electrocardiogram.

Electrocardiography is a popular procedure in clinical settings that is utilized for assessing heart rate, diagnosing any conduction problems, and checking heart function. Continuous ECG machines are applied in clinics and laboratories to study the performance of the heart for prolonged periods of time. These devices allow instant visualization and recording of physiological information when monitoring patients.

Real-Time ECG Monitoring System is a combination of sensing, signal processing, and display systems that makes it possible to continuously monitor the activity of the heart without any break. The system uses sensors to detect electrical signals from the body, performs signal processing through embedded hardware, and displays the waveform on the display system for viewing purposes.

3.2 Structure and Function of the Human Heart

The human heart is a muscle found within the chest cavity between the two lungs and tilted slightly to the left of the body. The heart is one of the most important organs of the circulatory system and its main function is to ensure there is a constant flow of blood to the entire body. This is achieved by the heart contracting and relaxing repeatedly, thus ensuring that oxygen and nutrients reach different parts of the body, while at the same time removing carbon dioxide and other metabolic wastes from the body.

Heart is comprised of four chambers working cohesively in the process of blood circulation. These chambers are differentiated by muscular barriers referred to as septum which ensure there is no mixing between oxygen-filled blood and deoxygenated blood.

The four chambers of the heart are:

- Right Atrium
- Left Atrium
- Right Ventricle
- Left Ventricle

Right atrium collects deoxygenated blood brought back by the body through the superior and inferior vena cava. This blood flows into the right ventricle which pumps it to the lungs using pulmonary artery for oxygenation. Blood containing oxygen will return to the left atrium via pulmonary veins after oxygenation in the lungs. From here, the left ventricle will pump oxygen-rich blood to the other body parts via aorta.

The synchronization of the flow of blood in the heart involves well-coordinated muscle contractions of each chamber. In this regard, the synchronization occurs through an electrochemical mechanism called the electrical conduction system, whereby electrical impulses are generated to control the contractions. The heart's electrical conduction system comprises specialized cardiac cells able to produce electrical impulses automatically without any stimulation from outside sources.

The following sections detail the major components of the heart's electrical conduction system.

Sinoatrial (SA) Node

The sinoatrial node comprises specialized cardiac cells forming a small group. They are found in the top wall of the right atrium and form the main source of heart's electrical impulses. The sinoatrial node is automatic in its impulses generation at fixed intervals of time hence creating a constant rhythm of contraction of the heart. When the impulses generated by the sinoatrial node spread in the atrial walls, they cause atrial muscle contraction and help move blood into the ventricles.

Atrioventricular (AV) Node

The Atrioventricular node is positioned in the region where the atria meet the ventricles. The main role of this node is to accept the impulses coming from the atria and control their passage to the ventricles. By causing delays in the transmission process, it gives enough time for the ventricles to fill up with blood before the process of contraction takes place. Such a mechanism ensures proper functioning of the heart by preventing the simultaneous contraction of both atria and ventricles.

Bundle of His

The Bundle of His is a system of fibers responsible for transporting the electrical impulses from the AV node to the ventricles. Splitting into right and left bundle branches running through the interventricular septum, the bundles transport impulses to the ventricles, preparing them for contractions.

Purkinje Fibers

Purkinje fibers are special fibers that help transmit the electrical impulses to all parts of the ventricles very quickly. As a result, there is uniform contraction of the ventricular walls and the heart can effectively pump blood to the systemic and pulmonary circulatory systems.

The correct functioning of the heart's conducting system plays a key role in ensuring that electrical impulses are produced in an orderly manner and delivered in the required order. The result of this is that the person will maintain a steady pulse rate, normal blood pressure, and constant blood circulation within his body. The process of ECG monitoring entails measuring the electrical impulses produced through this conducting process.

3.3 ECG Waveform and Its Components

The Electrocardiogram Waveform is an image showing the electrical impulse of the heart during one heart contraction. This impulse is produced when ions cross cell membranes of cardiac tissue, creating currents that run through the heart muscle. The ECG Waveform represents a chain of events in the process of heart chambers' contraction and relaxation. Different segments and waves seen in an ECG are specific for particular phases of cardiac activity and provide measurable data about heart beat rhythm and its conducting system.

Real-time electrocardiography allows for visual tracking of heart activity on the screen. Identifying segments of the waveform correctly helps assess the heart's work properly. Standardized ECG waveform contains certain segments and waves that can be seen in any normally working heart. These features will be described later.

3.3.1 P Wave

The P-wave is an indicator of the first atrial muscle activation by electric signals. This happens because the impulses emanating from the sinoatrial node diffuse through the atrial walls, making them ready for contraction and pumping of blood into the ventricles. Generally, the P-wave looks like a small upward wave on an ECG trace.

It is the measurement of the duration and amplitude of the P-wave that can show us some features of conduction in the atria. An ideal P-wave is supposed to be smooth in form and appear prior to the onset of ventricular activity.

3.3.2 PR Interval

The PR interval is the interval between the initiation of atrial activation and the commencement of ventricular activation. The PR interval is the time taken by the electric current to move from the atria through the atrioventricular node and to the ventricular conducting tissues. The PR interval highlights the synchronization of the activities of the atria and ventricles.

The PR interval is one of the essential components of an ECG because it demonstrates the efficiency of signal propagation in the heart. The constant value of the PR interval means that there is a controlled movement of the electric currents through the

heart conducting system.

3.3.3 QRS Complex

QRS complex is the electric activity associated with the contraction of ventricular myocardium. It happens because of the rapid spread of the electrical impulses from ventricles that cause these chambers to contract and discharge blood into pulmonary and systemic circulation. The QRS complex is the most conspicuous part of ECG because ventricular muscles produce stronger electrical impulses compared to those of atria.

It is composed of three waves, which include:

Q wave - first downward movement

R wave - sharp upward movement

S wave - downward movement after R wave

The duration of QRS complex is proportional to the velocity of ventricular conduction. Thus, the QRS complex gives evidence of normal ventricular conduction.

3.3.4 ST Segment

ST segment refers to the point when the ventricles have stopped contracting and are yet to undergo repolarization. At this stage, there is still electrical activity but no electrical change. The ST segment is depicted on the ECG graph by a straight line found between the QRS complex and T wave.

The location of the ST segment is crucial in determining the stability of the heart's muscles in terms of electrical activity. A stable ST segment shows that the heart muscles have stopped contracting and that it is ready for its next stage.

3.3.5 T Wave

The T wave refers to the electric relaxation of the ventricles after contraction. This period is termed repolarization since heart cells are being restored to their resting electric state. T waves are normally seen as gradual upward curves from the ST segments.

The analysis of T wave is essential in determining the condition of electric restoration of the ventricles. Watching for the T wave is important in ascertaining that the heart muscles have relaxed enough before another heartbeat takes place.

Components of ECG Waveform Importance in Monitoring

All components of the ECG waveform help achieve a comprehensive picture of cardiac electrophysiology. The analysis of the components allows one to monitor the cardiac rhythm, determine the conduction efficiency, and keep continuous control over heart signals.

ECG Monitoring System in Real-Time: The detection and visualization of components of the ECG waveform facilitate the recording of heart activity for the physiological signal monitoring process.

3.4 ECG Signal Characteristics

There are specific electrical characteristics of electrocardiogram (ECG) signals that will guide in determining how the signals are obtained, processed, and displayed in a monitoring system. The characteristics indicate the level, duration, and frequency nature of the electrical signals produced by the heart. It is important to know the electrical characteristics for the design of the hardware and software used in a Real-Time ECG Monitoring System. Knowledge of the characteristics of the signals will help in achieving accurate detection of heart activity and stable operation of the system.

Electrocardiogram signals are physiological signals. They are biological signals and are produced due to the movement of ions in the cardiac tissue. As the electric current travels across the heart, it causes the generation of voltage differences which are then picked on the surface of the skin. This voltage difference will be responsible for the waveform of the ECG signal. The nature of the signals is such that the magnitudes are very small and susceptible to external environments.

Features of ECG Signal

Amplitude of the Signal

Amplitude of the signal is basically a measure of the voltage or energy of the electric signal produced by the heart. Amplitude for the signal generated by the ECG is fairly low compared to other electrical signals that are commonly used in electronic circuits. Typically, ECG signal amplitude will lie within the range of millivolts.

Standard ECG signal amplitude readings include:

0.5 millivolts to 5 millivolts

The amplitude will depend on various factors like the positioning of the electrodes, skin conductance, and the electrical activity of the heart. Due to the small signal amplitude, there will have to be some form of amplification to raise the signal level for accurate processing.

ECG Signal Frequency Range

The ECG signal frequency range refers to how fast the electric signal varies. This means that each segment of the ECG waveform has a unique frequency range. In some cases, the variation in the electrical signal will be fast, for instance, the occurrence of the QRS wave. In other instances, it will vary slowly, for example, baseline shift.

The usual frequency range of the ECG signal is between:

0.05 Hz to 150 Hz

Adequate control of the frequency range helps in maintaining the integrity of the ECG signal wave. Filters are usually employed to filter frequencies that do not fall within the desired range. This helps to eliminate noise in the waveform.

Heart Rate Range

The heart rate refers to the number of heartbeats per minute. This factor can be considered an important indicator of the performance of the heart and the condition of

physiological functions. In the ECG monitoring system, the heart rate is measured through calculating the duration of intervals between successive heartbeats.

The standard range of heart rates of a healthy individual at rest is usually:

60 to 100 beats per minute

Measuring heart rate gives constant data on the condition of the heart and makes it possible to observe any changes in the heart rhythm. Monitoring systems show heart rate indicators together with the ECG wave.

Signal Duration and Timing

The signal duration refers to the period taken by the flow of electrical currents within different areas of the heart to reach their destination points. It indicates the amount of time each segment of the waveform takes depending on the speed at which the electric currents pass through various heart regions. The timing of each segment should be accurately measured.

This will help in detecting the order in which electrical activities occur during one heartbeat. Effective timing is essential to ensure proper synchronization of contractions in the atria and ventricles.

Noise Sensitivity and Interference

ECG signals are highly susceptible to any forms of noise and interferences because of their weak signals and nature of the biological signals. Electrical devices, movements, and the environment could all cause some form of noise that interferes with the accuracy of the ECG signals. Such noise could either change the waveform or decrease its clarity.

Interferences could come from:

- Power supply system
- Muscular movements
- Electromagnetic devices
- Loose electrode connection

For accurate and reliable signals, different techniques of signal conditioning are used in the monitoring system.

Importance of ECG Signal Characteristics for the Design of Systems

Knowledge of the signal characteristics is necessary to choose the right electronic components that would be used for the design of circuits that will monitor the signals.

In a Real-Time ECG Monitoring System, the use of proper signal characteristics is required for stability and efficient working of the system for observing the signal waveforms continuously.

3.5 Sources of Noise in ECG Signals

The electrical signals generated from the human heart are weak in amplitude and prone to interference from other environmental factors. An electrical disturbance which modifies or interferes with the original ECG waveform is termed noise. Noise can distort

the waveform, frequency, and magnitude of the signal, thus affecting the interpretation of the waveforms and the efficiency of the ECG monitoring process. Thus, determining the origin of noise and implementing ways to control it is a vital aspect in the process of designing ECG monitoring devices.

Noise can emanate from external electrical equipment, movement of the individual being monitored, and wrong connection of sensors. Disturbances arising from the above causes can lead to irregular waveforms, making it impossible to analyze the electrical activities of the heart. To ensure signal integrity, noise reduction measures such as filtering and effective grounding of electrodes are common in modern-day ECG monitoring systems.

Common sources of noise in ECG waves are listed below.

Power Line Interference

The most prevalent form of interference in biomedical signal acquisition is known as power line interference. The operation of an electrical system is governed by a constant frequency. The electrical frequency often used in many locations is 50 hertz. There is a possibility that the electrical frequency will get incorporated into the ECG circuitry via wires, electronic equipment, and any other electrical apparatus that might be operating nearby. Consequently, there could be a periodic wave that gets embedded in the ECG waveform.

Power line interference manifests itself as a continuous ripple pattern within the ECG signal. In order to mitigate power line interference, various measures can be taken within the ECG acquisition circuitry. Some of these include shielding of cables and proper grounding of equipment.

Noise due to Muscle Movement

Muscle movement noise arises due to the electrical activity that occurs during muscle contraction. As the individual makes any movement, changes posture, or feels muscular stress, there are new electrical impulses that are formed inside the body. They may interfere with the ECG signal since they both arise as a result of electrical activity in the tissues of the body.

Such noise is often found when an individual moves around, breathes deeply, or makes sudden changes in posture. Muscle movement noise may cause spikes or quick changes in the ECG trace. It becomes necessary to maintain a constant body position during the process of recording and ensure proper electrode fixation.

Baseline Wander

Baseline wander is characterized by slow and continual change in the level of the signal that appears in the ECG recording. Rather than maintaining a constant level, the baseline itself slowly changes to either a higher or lower level. This causes problems with discerning wave components from the signal, particularly for signals recorded over an extended period of time.

The reason for baseline wander is due to the motion created from processes such as breathing or physical movement as well as gradual changes in the placement of the electrodes relative to the skin surface. This creates slight variations in the electrical connections between the body and the electrodes. High-pass filters are often utilized to remove low frequency drift.

Noise Due to Electrode Contacts

The electrode contacts create noise if there is an incorrect electric connection between the surface of the electrode and the skin surface. The reasons for poor contact include dry skin, loosely attached electrodes, low quantity of conductive gel, and movement of the electrodes while recording the signal. When the electrical connection is not stable, the signal shows irregularities.

Good electrode connection is critical for proper signal measurement. The cleanliness of the skin surface, application of conductive gel, and firm placement of electrodes contribute to stable electrode contact..

Motion Artifacts Noise

Motion artifacts noise is generated due to physical movement of the electrodes or the wires attached to them. Such movement causes momentary fluctuations in the electrical resistance leading to irregularities in the ECG signal. The effect of this noise appears as a spike in the waveform.

Such kinds of noise typically occur if the cables are moved, twisted, or improperly connected.

Noise Filtering in ECG Measurements

Noise reduction in ECG recordings is an integral part of the process that helps ensure that the measurements remain accurate. Without noise filtering, clear waves can be observed and analyzed, which enables recognition of events related to the heart functioning and allows for continuous tracking of these events. In a Real-Time ECG Monitoring System, noise can be efficiently filtered using different approaches.

3.6 Signal Conditioning Techniques

The signal conditioning process plays a crucial role in the ECG monitoring system. It enhances the quality of the electrical signal derived from the electrodes before it undergoes further measurements and analyses. In most cases, the signal obtained from the human body is very weak, and at times, it contains some disturbances arising from both environmental and physiological factors. Therefore, the need for signal conditioning arises to improve its quality by enhancing the desired signal and eliminating any interferences.

Within a Real-Time ECG Monitoring System, signal conditioning circuitry is designed such that it retains the original wave shape features and causes minimal distortion. The circuit consists of amplifiers, filters, and converters, all of which combine to provide an effective ECG signal.

The main signal conditioning techniques used in ECG monitoring systems are described below.

Amplification

Amplification is the procedure whereby the ECG signal is strengthened in order for its processing and measurement through the use of electronic equipment to be feasible. The electrical signals produced by the human heart are normally characterized by very low levels of magnitude. As a result of their low level of magnitude, they require to be amplified before further processing.

In order for accurate measurements and analysis to be obtained, instrumentation amplifiers are normally applied in ECG systems. Such type of amplifiers are able to offer exact amplification and signal stability at the same time. Instrumentation amplifiers are also characterized by high level of input impedance and low output impedance. As such, high input impedance makes sure that minimum amount of current is drawn from the input signal. On the other hand, low output impedance enables efficient transmission of signals to the succeeding part of the circuitry.

Instrumentation amplifiers offer differential amplification, whereby the difference between two input signals is amplified while any other noise signals common to both inputs are rejected.

Filtering

Filtering involves the elimination of unnecessary frequency components of the ECG signal, but the useful frequency components remain in place. Filtering is important since ECG signals are prone to being corrupted by noise that originates from different sources. Filters refer to circuits that can allow some frequencies to be filtered while stopping others.

Different filters are employed in ECG signal conditioning systems depending on the types of noises found in the ECG signals.

Low-Pass Filter

This type of filter allows low-frequency signals to pass through while filtering the high-frequency noises in the signal. The source of high-frequency noise is usually electronic equipment, switching circuits, or electromagnetic interference.

High-Pass Filter

This filter allows higher-frequency signals to pass while filtering the very low-frequency components of the ECG signal. The source of low-frequency noises includes baseline wander or breathing movement or slowly changing electrode contact.

Band-Pass Filter

The band-pass filter transmits signals with frequencies within a defined range, whereas signals outside this range are not transmitted. The band-pass filter is an amalgamation of the high-pass and low-pass filters. In the context of ECG monitoring, the band-pass filter serves to allow only the frequencies that relate to heart activity while suppressing frequencies outside this range.

Notch Filter

A notch filter acts to suppress a certain frequency band in a signal. In ECG, the notch filter is generally applied to suppress any interference in the system generated by the operation of the electrical power supply at its nominal frequency.

Conversion from Analog to Digital Form

Once the ECG has been amplified and filtered, the resulting signal will still exist in analog format. For the purpose of further digital analysis and monitoring, it needs to be digitized. This is done through a specialized device called an Analog to Digital Converter or an ADC.

An Analog to Digital Converter takes periodic measurements of the analog signal and represents them in numeric form. Sampling frequency defines how often the measurement is taken while the resolution determines the degree of details preserved in the digital signal. Increased sampling frequency and resolution ensure increased accuracy of digitization.

Once the analog ECG signal has been converted into its digital form, it can be further processed by a microcontroller or similar device. The digital signal can now be saved into storage, transmitted elsewhere or viewed on a display. This makes it possible to monitor the functioning of a patient's heart in real time.

4 METHODOLOGY

4.1 Introduction

System Methodology in the Real-Time ECG Monitoring System refers to the methodical steps taken in designing, implementing, and running the ECG monitoring system. The methodology outlines how the different phases in acquiring the electric signals, processing those signals with electronic components, and displaying the processed waveform are performed. This helps ensure that the processes carried out in the system follow a proper methodology in order to be successful.

In the ECG monitoring system methodology, biomedical sensing technologies, signal conditioning circuits, microcontroller processing, and output display are used together to monitor cardiac activity in real time. Different phases in the methodology represent various functions that are performed in the system from one end of its operation to the other. This includes signal acquisition at one end using electrodes and signal display at another end using an interface display mechanism.

Appropriate methodology allows one to obtain signals accurately and process them correctly. The right choice and appropriate setup of system elements will allow obtaining the ECG signal without any distortions and processing the data effectively. Additionally, there are tests included into the methodology for verifying whether the system works properly in normal operational conditions. This makes sure that the system performance is consistent enough, which helps monitor heart activity safely.

The whole implementation process is based on stepwise design that consists of such stages as hardware design, software programming, signal processing, and system execution. At each stage, one has to follow all necessary technical specifications to be able to operate the system smoothly and obtain results reliably. Constant monitoring becomes possible because of the cooperation between different elements of the system.

4.2 System Architecture

The architecture of the Real-Time ECG Monitoring System consists of the systematic arrangement of hardware modules in the monitoring system along with their interactions. In simple words, the architecture of the system explains the manner in which the various components of the system are interconnected and the way in which the signal passes through different parts of the system until the end point.

The ECG monitoring system consists of a number of interconnected modules that work together as an integrated system to achieve the monitoring of cardiac signals in real time. Different modules are designated for specific tasks in the system and the appropriate interaction between these modules ensures the efficient operation of the monitoring system.

The following are the major modules of the ECG monitoring system.

ECG Sensor Module

It is the ECG sensor module that should detect electric potentials generated by the work of the human heart. It is considered the main input part of the system for monitoring the functioning of the heart. For its successful work, it needs to be provided with signals from electrodes applied to the human body at special measuring sites.

They collect information about the electrical potential generated by the work of the heart muscle and pass it to the circuit of the sensor module. It processes the collected information and turns it into a specific analog signal. It should operate reliably and efficiently because otherwise, no correct signals will be detected.

Signal Conditioning Circuit

The signal conditioning circuit helps improve the raw ECG signal to make it ready for processing. The signal that is detected by the sensor module is usually very weak and might have some disturbance due to external factors. The signal conditioning circuit increases the signal amplitude and reduces noise before transmitting the signal to the processing circuit.

The signal conditioning circuit consists of an amplification stage that increases the signal amplitude and a filter stage that eliminates unwanted noise. Proper signal conditioning is essential for preserving the waveform of the ECG signal to allow accurate processing.

Microcontroller Unit

The MCU is responsible for acting as the processing and controlling element of the ECG measurement process. In this case, it will receive conditioned analog signals from the signal conditioner and convert them into digital data through an embedded ADC module. Processing of this digital data will be done by the MCU based on predefined software code.

The MCU is tasked with ensuring that the system time is managed effectively, data is received and processed accordingly, and communication between the various modules is performed smoothly. Some of the signal processing operations performed by the MCU include filtering, data communication with the display unit, and waveform generation.

Display Unit

The role of the display unit is to show the output of the ECG signal that has undergone processing in a visual way so that it can be easily monitored. It takes input from the microcontroller in digital form and converts it into waveforms that indicate heart activity. It constantly displays the waveforms based on newly acquired data.

Other than displaying the waveforms, the display unit can also show the status of the signal and heart rate values. Proper display of the ECG signal enables easy monitoring of heart activity.

Power Supply Unit

The power supply module supplies the necessary electrical energy for running the different parts of the Real-Time ECG Monitoring System. The module converts the input electrical power to an appropriate voltage level for use within the electronic devices. Reliable operation of the system requires a consistent supply of power to avoid distortion of the signals due to fluctuating voltages.

Proper power management allows the system to run safely and prevents damage to electronic parts due to inconsistent levels of power.

In addition to that, working together of the modules discussed above creates the full system architecture needed for continuous acquisition, processing, and display of the ECG signals. This makes the Real-Time ECG Monitoring System able to monitor the heart activity accurately.

4.3 ECG Signal Acquisition

Stage one of the monitoring process entails receiving bioelectrical signals produced by the heart through electrodes.

Electrodes are attached to the skin to receive bioelectrical signals. These signals are then sent to the signal conditioning unit where more processing is done.

Electrode attachment is crucial for successful signal reception.

4.4 Signal Processing

The process of signal processing includes enhancing the quality of the signal and obtaining information from the ECG signal.

The processes involved include:

Amplification

Filtering

Conversion of analog signal to digital

Peak detection

Calculation of heart rate

4.5 Hardware Design

The hardware design is based on choosing components that will ensure proper performance and compatibility.

Components of the hardware system are as follows:

ECG sensor module

Microcontroller

Display unit

Power supply

4.6 Software Design

The software handles all the operation within the system and manages the ECG signal processing. The functions of the software include:

- Data reading from sensors

- Noise filtering

- Waveform peak detection

- Heart rate calculation

- Displaying the ECG waveforms

This software is programmed using embedded programming techniques.

5 COMPONENTS USED

Components used:-

5.1 Hardware:-

5.1.1 ESP8266:-

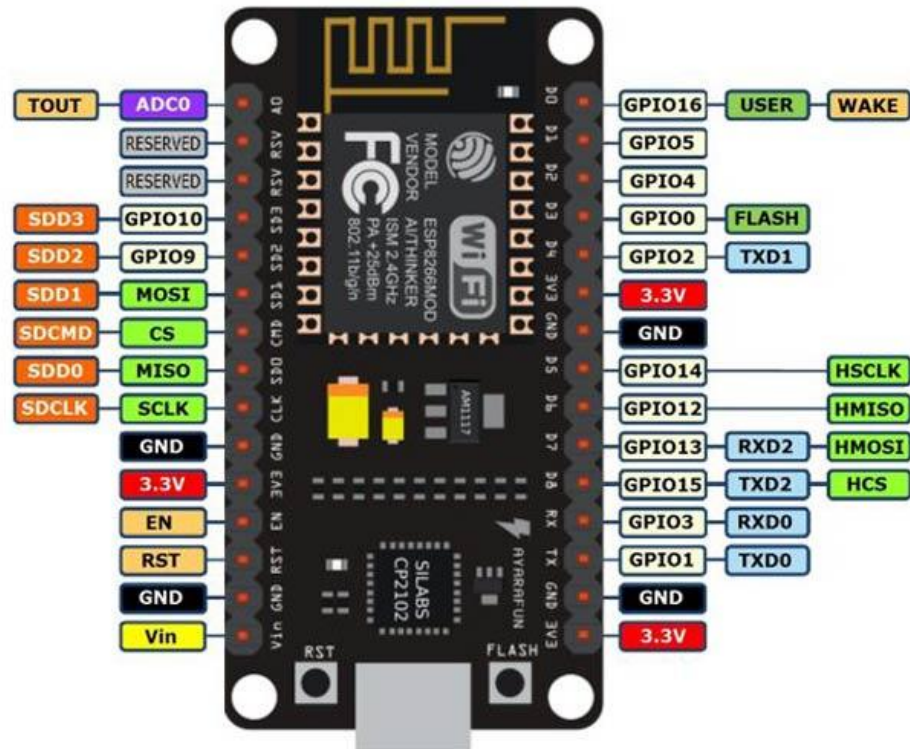


Figure 5.1.1: ESP8266

ESP8266 is a Wi-Fi-based microcontroller board that is extensively used for developing Internet-of-Things based applications. The board is manufactured by Espressif Systems and features a TCP/IP stack on board which allows devices to connect to the internet without the need for any separate processor. ESP8266 is not only able to act as a microcontroller but it can be used as a Wi-Fi module along with other microcontroller boards like Arduino. It can handle a variety of protocols including HTTP, MQTT, and TCP/UDP. This makes it very useful for building systems like a smart home system and wireless sensor network.

5.1.2 ECG disposable Ag/AgCl electrodes

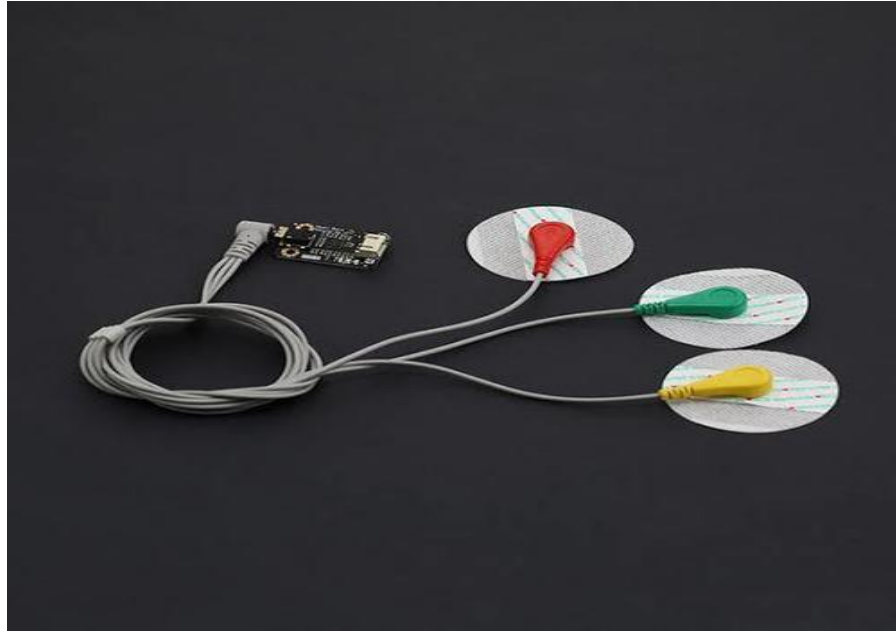


Figure 5.1.2: ECG electrodes

Electrocardiograph (ECG) disposable Ag/AgCl electrodes have become common in biomedical sensing because of their dependable electrical properties. Electrodes of this type utilize a silver and silver chloride connection and thus exhibit low impedance against the skin and reduce noise while sensing the signals. In addition, being disposable in nature, they ensure hygiene and are not prone to contamination when used on different patients hence making them applicable in both laboratory and mobile applications. They come already gelled to promote contact between the electrode and the skin hence producing reliable results. Because of their reliability, consistency, and easy application, Ag/AgCl electrodes are often employed in electrocardiograph monitoring systems.

5.1.3 Jumper wire

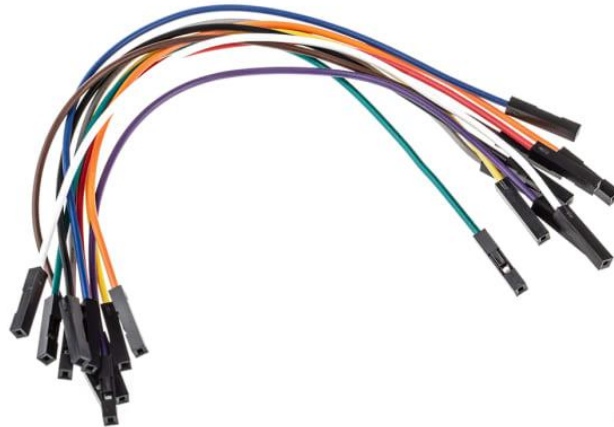


Figure 5.1.3: Jumper wire

Jumpers are basic connectors that are utilized to create temporary interconnections between two circuit components by not requiring soldering. Jumpers are widely applied in the stages of testing and prototyping, particularly for circuits on breadboards and boards such as Arduino or microcontroller boards. Jumpers can be categorized into three types, including male-to-male, male-to-female, and female-to-female jumpers. They are insulated and come in various colors and lengths for convenient identification. In devices such as ECG monitoring system, jumpers are utilized in order to ensure appropriate signals between components within the circuit.

5.1.4 Bread board

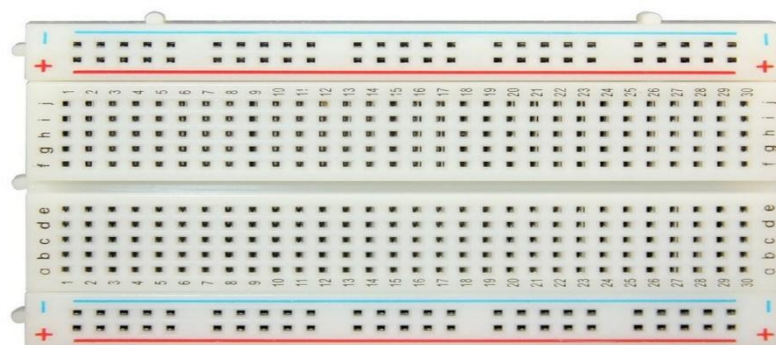


Figure 5.1.4: Bread Board

A breadboard is a reusable device that facilitates prototyping electronic circuits without needing to use soldering techniques. It is made up of a matrix of holes that are connected by metal strips, making it easier to insert resistors, integrated circuits, and wires

to make a connection. It provides predetermined paths of electrical connectivity, including power lines, where voltage and ground can be delivered. Breadboards are very common in the designing and testing of electronic circuits due to their simplicity in changing and debugging the circuits. They can be applied in applications such as the heart rate monitor where they facilitate assembly and checking of connections among sensors, microcontrollers, and other electronic devices.

5.1.5 Ads8232 module

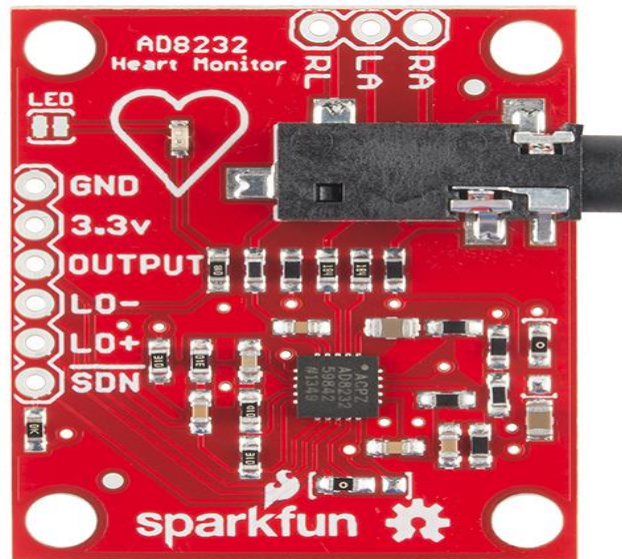


Figure 5.1.5 Ads8232 module

The AD8232 module is a compact signal conditioning module made up of the AD8232 chip by Analog Devices. This chip has been specially designed for use in biomedical signal measurement. In addition to that, the module contains amplifiers, filters, and noise suppressers that help in isolating clean ECG signals despite the motion artifacts and electrical interference. The module uses Ag/AgCl electrode leads to monitor electrical activities of the heart, and it gives out analog output signals which can be interfaced to any microcontroller such as Arduino. It is very popular due to its small size, energy-saving ability, and reliability.

5.1.6 USB cable:-



Figure 5.1.6 USB Cable

A USB cable can be used to create a link between electronic devices for power and data transmission purposes. USB cables can be made of standard connectors such as USB-A, USB-B, Micro-USB, or USB-C based on the devices that are being connected.

For instance, a USB cable may be needed to connect the microcontroller or the

development board to a computer during the design of an ECG monitoring system. The use of a USB cable will enable easy programming, data transfer, and power distribution.

5.2 Software:-

5.2.1 Arduino ide serial plotter

The Serial Plotter in Arduino IDE is an integrated plotting software that allows visualizing the data being generated by a microcontroller. The Serial Plotter plots real-time data from the input values obtained from the serial port as waveforms, and thus, it helps analyze various signals such as ECG signals, temperature fluctuations, and sensor outputs. Unlike the Serial Monitor which displays the received input values in text format, the Serial Plotter represents the incoming data as real-time graphs, enabling easy interpretation of the data being analyzed. The Serial Plotter is used in ECG monitoring devices where one can view the waveforms being created during device operation.

6 IMPLEMENTATION AND EXECUTION

6.1 Introduction

The implementation process involved in the Real-Time ECG Monitoring System is the physical building process of the device through assembly of the hardware and programming the software necessary for operation of the system. In this stage, the circuits that had been designed will be assembled and electrical connections set up. The microcontroller will also be programmed to undertake the processes of acquiring, processing, and displaying ECG signals in real time.

The implementation involves the use of electrode attachments, ECG sensor module configuration, design of signal conditioning circuits, and programming of microcontroller unit. All the components of the hardware are wired using the circuit diagram in order to facilitate proper flow of signals from the sensing stage through to the output display. Proper wiring of the components, correct positioning of the components, and correct voltage application are ensured to provide for stable operation.

Development of software is vital in implementing the system. The software for the microcontroller is written so that it helps in controlling the functions of the system by receiving analog inputs from the ECG sensor, processing the input to convert it from analog form to digital using the ADC, processing the data, and finally sending out the processed data through the output interface. It also takes care of the timing functions in order to achieve proper sampling of the signal and generation of waveforms.

Once the hardware assembly is completed and the software code has been uploaded, the system is then run to ensure that it works correctly. Various tests are conducted to ensure that the ECG signal is received without distortion and that it is shown in real time continuously. This shows that the various modules work together as one unit.

6.2 Hardware Implementation

Hardware implementation entails putting together and linking the various components of the system to develop an efficient and functional monitoring machine based on the Real-Time ECG Monitoring Circuit Design. The proper wiring of each unit ensures that there is effective communication between the units. Signal integrity can be achieved by ensuring that all hardware connections are properly done without disrupting the functioning of the whole ECG Monitoring Machine.

The process of hardware implementation consists of putting together all the necessary components in developing an effective monitoring machine. The connection of the components is systematically made to ensure the continuity of signals and prevent any electrical failures of the machine's components.

The hardware implementation process entails the following stages.

Connecting ECG Electrodes to the Sensor Module

Electrodes are placed on specific points of the human body to receive electrical signals created due to the functions of the heart. These electrodes are then coupled with the input ports of the ECG sensing module through the help of conducting wires. Such

coupling is done in order to ensure that the detected electrical signals from the body are relayed precisely to the sensing circuit.

Connecting the Sensor Module to the Microcontroller

The ECG signal output from the ECG sensor module goes to the analog input port of the microcontroller. This helps the microcontroller capture the ECG signal for subsequent processing. The signal cables are joined using insulated wires to ensure there is no interference with the signals. A proper link between the ECG sensor and the microcontroller is crucial for successful data collection.

Connecting the Display Unit

The display unit is attached to the microcontroller using a communication channel to provide a visual representation of the signal after processing of the ECG signal. Through this connection, data will be transmitted from the microcontroller to the display unit in the form of digital information. Data will be continually received by the display and converted into graphics for the ECG waveform.

Providing Power Supply to the System

The regulated power supply is also linked to all the hardware modules to facilitate the supply of the necessary electrical energy. The power supply ensures that there is a constant voltage level within the sensor, microcontroller, and display modules. The regulation of the voltage levels and grounding help to ensure that the voltage levels remain constant throughout the process.

Assembly of the hardware and connecting the electrical circuits effectively enhance the stability of the system, thereby reducing any interference of signals and helping monitor heart activities.

6.3 Software Implementation

The software part of the Real-Time ECG Monitoring System will be used to control the system operations, hardware modules and for the processing of signals needed in continuous monitoring of cardiac signals. Software program will be used to coordinate activities in the process of receiving signals from the ECG sensor, conversion of the analog signal received to digital signal, analysis of the acquired signals, and finally displaying results on the display unit.

The software design uses embedded software programming approach that allows the system to handle real time data effectively. The program is divided into functional modules that are responsible for performing particular operations like initialization of the system, data acquisition, analysis, processing, and display of results among others. Proper software design will guarantee accurate measurement of the signal, proper timing of the process and effective monitoring operation.

The following are the software operations involved.

Initializing Hardware Components

The initialization of hardware begins with the setup of all the hardware components interfaced to the microcontroller. The initializations include configuration of I/O ports, communication interface, as well as the activation of any internal peripherals needed in

the system operation. The analog-to-digital converter is initialized to accept inputs from the ECG sensor. Also, timers must be initialized to maintain a constant sampling rate throughout the acquisition process.

Reading ECG Data from the Sensor

The analog readings from the sensor module are read by the program constantly through the input interface of the microcontroller. The analog readings from the sensors are then converted into digital form through the analog to digital converter, which makes the reading ready for analysis by the computer. The readings are taken periodically to facilitate constant monitoring of the heart.

Processing Signals

Once the data is obtained using the digital ECG method, the signal processing functions take place where signal processing is carried out to ensure that the signals are of high quality and that the data can be analyzed effectively. The processes include removal of any form of noise, establishing stability in the signals, and formatting the signals.

Calculating Heart Rate

The heart rate is calculated based on the period between consecutive peak signals from the ECG signal waves. The period between consecutive heart signals is measured and converted to beats per minute. The conversion of this measurement gives a numerical description of the heart activities.

Displaying ECG Waveform

The acquired processed ECG signal and the calculated heart rate data are sent to the display unit for graphical representation. The software provides continuous update of the display as soon as new information is available. It ensures that the ECG graph can be seen on the display screen in real time.

The programming language used in this software is Embedded C, and it is executed by the microcontroller.

6.4 System Testing

Testing of the system is done to determine its operational efficiency, accuracy, and reliability after completing the assembly of the hardware and software design of the Real-Time ECG Monitoring System. The testing process ensures that all the components in the system work properly during the normal operation of the system. It also ensures that the system can monitor heart activities without fail.

The testing process includes analyzing the individual functionalities of the system and its overall functionality as a whole system. The modules are analyzed during operation to ascertain whether the hardware components communicate well with each other and whether the software program works. Regular results from the tests indicate that the system performs consistently.

System testing of the system includes the following activities.

Checking Signal Acquisition

Testing for signal acquisition is done to prove that the sensor module of the ECG sensor is capable of detecting electrical signals from the body. This involves checking the

electrode and sensor contacts to ensure proper signal acquisition. The output signal from the sensor is analyzed to ensure that there is consistent signal transmission to the microcontroller. Successful signal acquisition means that the sensor is operating efficiently.

Verifying Waveform Display

Waveform Display Testing

Testing is carried out to ascertain whether the outputted ECG signal is effectively depicted on the screen. The display screen is examined to confirm whether the waveform is continuously shown and changes in real-time with data input. The pattern and flow of the waveform are examined to ascertain that the outputted wave is not distorted or paused at any point in time.

Measuring Heart Rate Accuracy

The accuracy test for the heart rate is conducted to confirm that the heart rate value can be determined accurately from the ECG signals obtained. The measured heart rate is monitored for some time to ensure that the reading is steady. If the heart rate is determined accurately, then it means that the signals have been processed properly and timing is accurate in the code.

Observing System Stability

The test for system stability is carried out to determine whether the ECG monitoring system is capable of running without any failure for a long period. The system is energized, and then it runs for a set amount of time, with the output signal being observed for consistency. Running without any failures guarantees the proper functioning of the hardware and software used.

System testing ensures that all modules function well and proves the efficient working of the Real-Time ECG Monitoring System when monitoring cardiac activities continuously..

7 EXPERIMENTAL RESULTS AND COMPARISON

7.1 Introduction

Experimental results will be obtained by running the Real-Time ECG Monitoring System in an appropriate test setting, whereby the system's input variables and operating conditions will be controlled. At this stage, the system will be switched on, interfaced with its components, and left to operate continuously as ECG signals are captured. The observed results give measurable data concerning the system performance at real-time.

Experimental analysis is intended to measure the performance of the ECG monitoring system through the accuracy of signal detection, processing, and displaying operations. The measured ECG waveforms and heart rates will be evaluated to ensure that the system operates within its expected performance criteria based on the design. Reliable results show proper functioning of the hardware components as well as interaction with the embedded software application.

The process of evaluating the system performance involves a comparison of the output signals against the operational features of an ECG monitoring system. Critical performance variables like signal clarity, response speed, consistency in measurement, and stability are among the key factors to consider when evaluating performance.

The analysis of experimental results clearly proves the ability of the system to track the activity of the heart. Evaluation guarantees that the designed Real-Time ECG Monitoring System meets all the necessary conditions to function effectively and accurately monitor the heart signal.

7.2 ECG Waveform Observation

The Real-Time ECG Monitoring System is responsible for capturing the electrical signals from the heart and displaying the waveform associated with the detected electrical signals. The waveform displayed is the result of the electrical activity taking place in the heart during each cycle. It can be continuously monitored as proof of the ability of the system to capture, process, and display ECG signals in real time.

The ECG waveform consists of specific parts that represent the heart activity at various stages. All the components of the waveform are observable on the screen, showing that the electrical activity of the heart is properly captured, processed, and displayed by the system. The signal is stable throughout the operation, confirming proper functioning.

The components of the displayed waveform include.

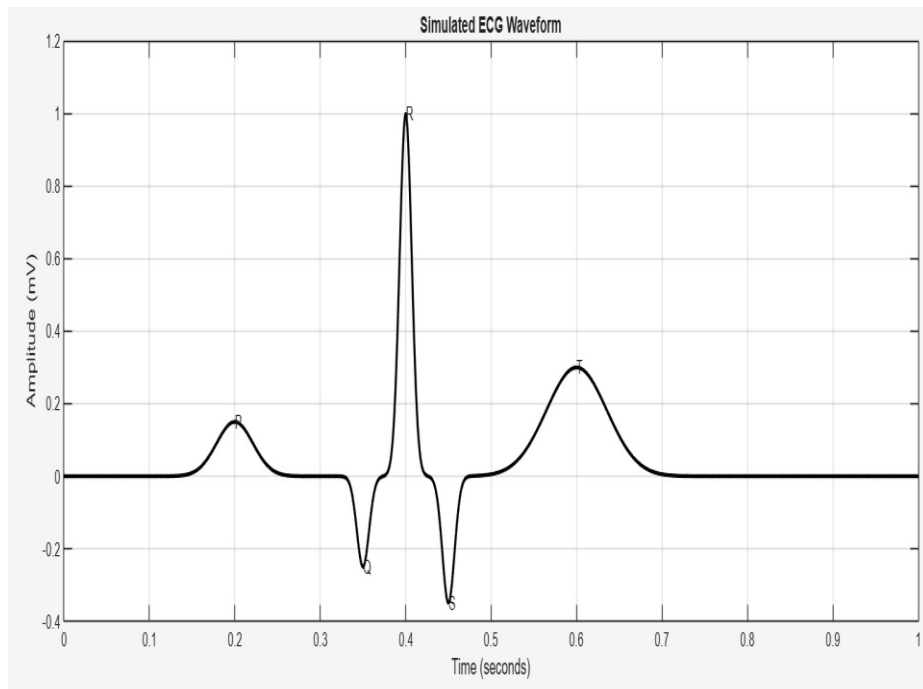


Figure 7.2.1 ECG Waveform

P Wave

This wave can be seen as a small, positive deflection on the ECG waveform and reflects the electrical activity of the muscles of the atria. It occurs at the start of each heartbeat, indicating that the atria have started contracting. Its presence signifies correct sensing of the electrical activity of the atria.

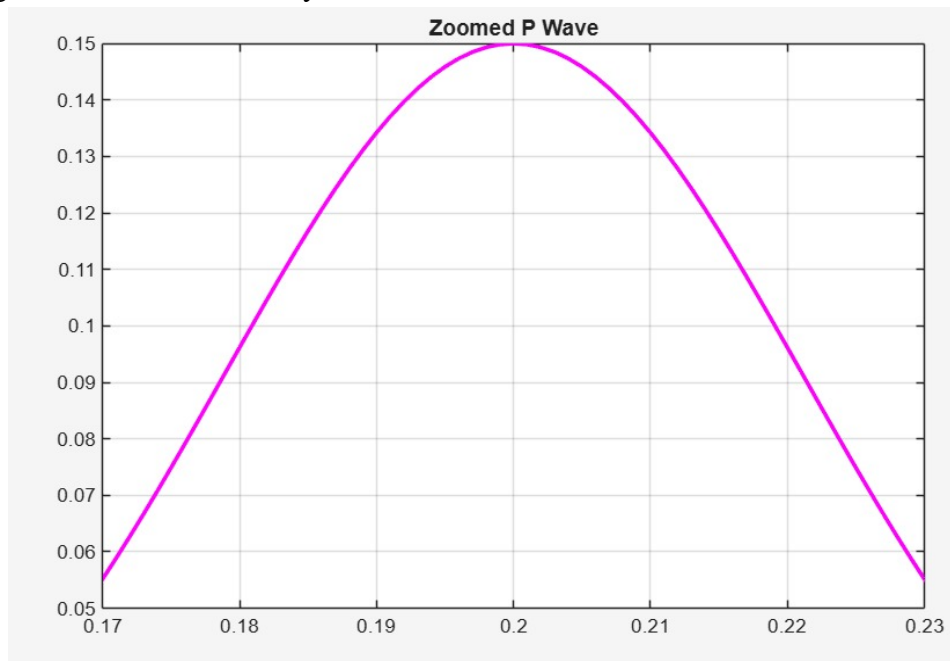


Figure 7.2.2 P Waveform

QRS Complex

This can be seen as a sharp part of the wave which is more distinct with greater amplitude than other parts of the waveform. This represents the activation of the ventricular muscle fibers and is associated with the contraction of the ventricles. It is detected consistently throughout each beat of the heart.

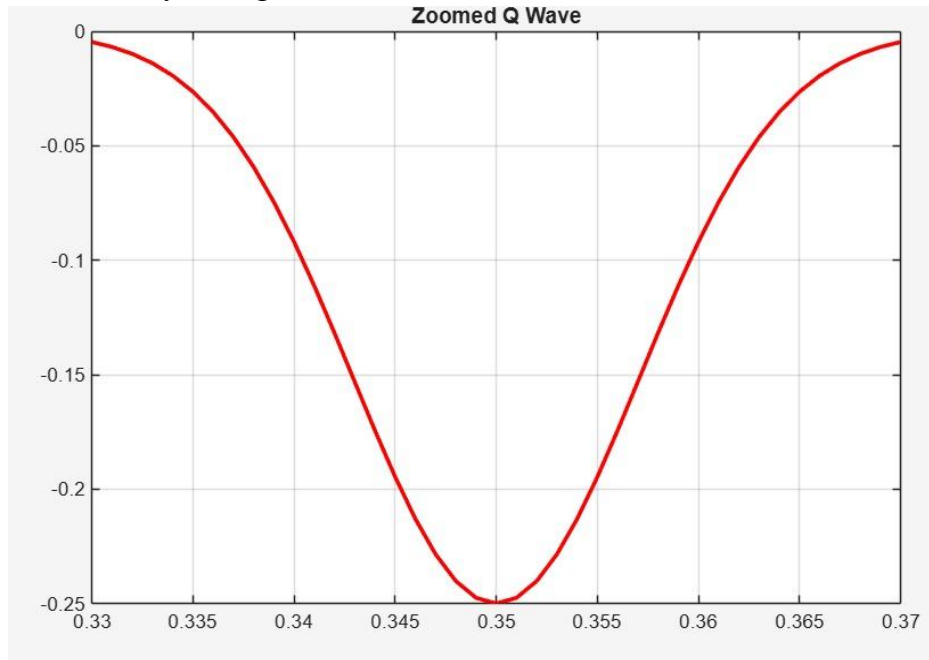


Figure 7.2.3 Q Waveform

T Wave

The T-wave follows the QRS complex as a smooth and rounded waveform component. This reflects the re-polarization phase where there is a recovery from ventricular depolarization. The distinct appearance of the T wave shows that the signal acquisition is adequate for reliable results in the monitoring period.

The waveform pattern is consistent during prolonged observation times. Prolonged observation proves that the Real-Time ECG Monitoring System can sustain stable and consistent signal acquisition and waveform presentation.

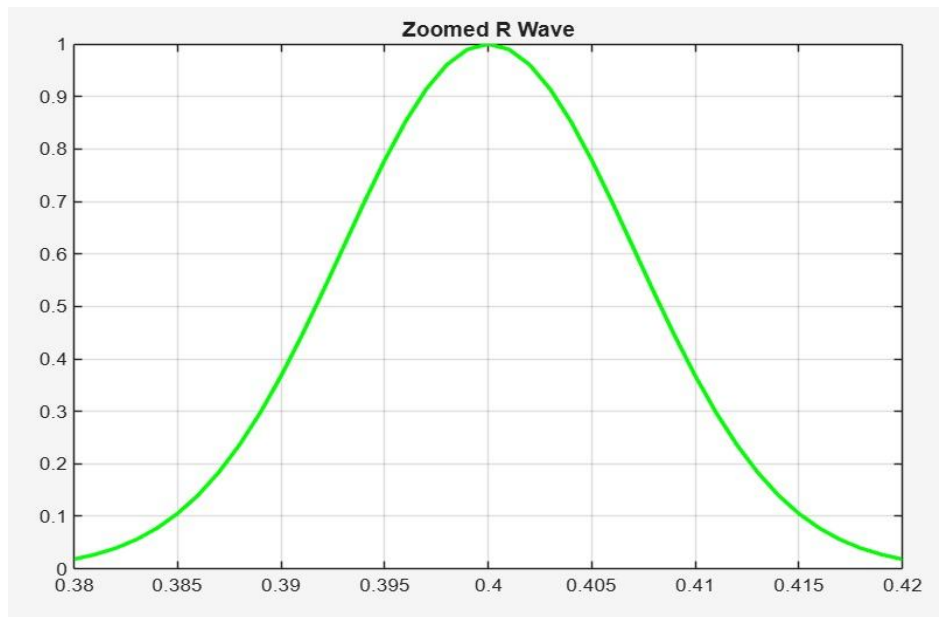


Figure 7.2.4 R Waveform

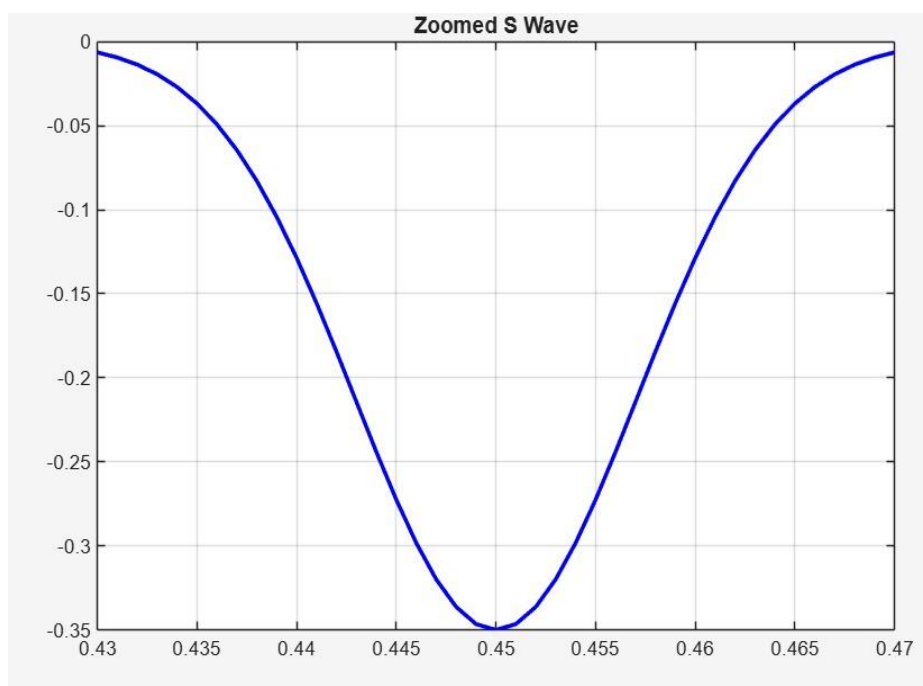


Figure 7.2.5 S Waveform

7.3 Heart Rate Measurement

Heart rate is determined using the Real-Time ECG Monitoring System based on the analysis of the electrical signal recorded from the ECG waveform generated through the system. Measurement of the heart rate involves the use of the R-peaks, which are the highest points in the QRS complex. Each heartbeat produces one R-peak; therefore, the peaks serve as a reliable reference for determining the number of heartbeats within a

predetermined period. The peaks are measured continuously to ensure the accuracy and real-time determination of heart rate.

The heart rate value is presented in numerical form together with the ECG waveform displayed graphically. This display makes it possible to view the electrical activity of the heart and its heart rate simultaneously. The values on the display are continuously updated using newly acquired ECG values through the microcontroller.

The heart rate determination process involves measuring the difference in time between two consecutive R-peaks in an ECG signal. The microcontroller measures the time interval between two R-peaks and presents the results in beats per minute.

Throughout the monitoring process, the values for heart rates remain constant when there is stability in the ECG signal. The values measured during this process can be seen to be constant over prolonged monitoring processes. This shows that the signals being measured are correct and that the algorithm has worked properly in processing. The consistency of heart rate measurement implies successful signal analysis by the RT ECG MS.

7.4 System Performance Evaluation

Performance Evaluation

Performance evaluation of the Real-Time ECG Monitoring System can be carried out through the study of the system's operations during its constant functioning while determining the quality of the signals that it generates when in use. Performance evaluation should be carried out after successful testing of the system through experimentation. Performance evaluation of the Real-Time ECG Monitoring System will involve an analysis of the ability of the system to capture, process, and display information in a reliable way.

Performance criteria for system evaluation

System performance can be determined using various criteria, which are used to evaluate the reliability of the system operations during use. These criteria help in determining whether the system performs efficiently in ECG monitoring and provide measurable parameters for evaluating its performance.

The system performance is evaluated based on the following parameters.

Signal Clarity

The clarity of the signal depends on how good the ECG waveform looks when it is shown on the display screen. If the signal is clear, it implies that there was successful filtering of noise by the system, as well as retention of the essential components of the waveform of the ECG signal. Clarity of the waveform can be seen to check whether its shape is smooth and distinguishable for proper identification of the waveform components.

Measurement Accuracy

Accuracy in measurement refers to the precision with which the values for the heart rate generated by the system based on the detected ECG waves are accurate. The generated values are checked continuously while the system is in use to ensure that the values are consistent and match the heartbeat wave detections.

System Stability

The stability of the system is described by the continuous performance of the ECG monitoring system without any interruption, loss of signals, or damage to any part of the hardware. The system is monitored for an extended period to ensure that there is consistency in the output of the heart waveforms and the heart rate.

Continuous Operation

Continuous operation implies the ability of the system to monitor heart activity without any interruptions for the long period of time. The system is powered and allowed to function by acquiring signals, processing and displaying them properly. The stable operation of the system without performance degradation is evidence of efficient power supply and proper usage of resources in the monitoring system.

The obtained data indicate the stable operation of the Real-Time ECG Monitoring System under any conditions considered.

8 DISCUSSION AND ANALYSIS

8.1 Introduction

The current chapter provides an analysis of the Real-Time ECG Monitoring System in light of the observations made during the process of its testing and implementation. This analysis focuses on the operations and performance of the developed system during the stages of acquiring, processing, and visualization of real-time ECG signals generated by the subject.

Particular attention will be paid to the consistency of ECG signal acquisition by the use of electrodes and analog circuitry, as well as to the efficiency of the noise-filtering procedures implemented for the sake of enhancing the quality of the raw signal. In addition, the chapter analyzes the performance of the system in terms of heartbeat detection and creation of ECG waveforms in real time at the display interface.

In addition, the chapter presents the performance of the system in terms of the quality of the signal, processing speed, and continuous monitoring capabilities when the system operates normally. The results gathered show that the system successfully works together with its hardware and software parts to continuously monitor the ECG signal and calculate the heart rate. This chapter is the basis of the Real-Time ECG Monitoring System operations.

8.2 System Performance and Processing Efficiency

The efficiency of the Real-Time ECG Monitoring System was assessed by evaluating the efficiency of the system in handling continuous ECG signals. It was evident that the system was able to acquire bioelectric signals via the electrodes and convert the acquired signals into digital signals through the use of the signal conditioning circuit and analog-to-digital conversion process. The sampling process was stable throughout the duration of the test run, thus acquiring data without interruption.

In the process of system operation, the microcontroller performed several functions simultaneously, including amplifying the signal strength, filtering, generating the waveforms, and communicating the data. The filter function of the microcontroller was efficient in minimizing the interference effects of the external electrical signals on the baseline and the acquired signals. This allowed the system to generate a steady waveform output for the observed cardiac activities. On the other hand, the processing function of the microcontroller involved processing the input signals according to the instructions provided.

Thus, the processing module was able to provide timely heart rate readings and waveforms to the display monitor. The communication process was smooth and reliable without any significant lag or communication errors during the test runs.

Moreover, the voltage regulation circuitry helped ensure equal voltage values for all the modules used by the circuitry, ensuring continuous operation without any shut-downs or restarts of the system. It can thus be observed that the overall efficiency and effectiveness of the Real-Time ECG Monitoring System was ensured in real-time operations.

8.3 Signal Integrity and Noise Filtration

The effectiveness of the ECG wave output in the Real-Time ECG Monitoring System is achieved through successful execution of the signal processing and filtering processes within the system. In the hardware section of the device, there is an instrumentation amplifier that increases the weak input signals from the body and ensures minimal interferences from other external electrical appliances. The amplifying circuit was able to provide stable signal amplification for the entire process, thus enabling successful ECG wave output.

Further filtering of the signal was undertaken using additional filtering devices. The analog filter circuits managed to minimize the variations that may occur due to the influence of muscle activities, power line fluctuations, and environmental noise. Consequently, the ECG wave output appeared smooth and consistent.

In terms of software, digital signal processing algorithms were implemented to further refine and stabilize the acquired signal after digitization. The filtering functions embedded in the program were constantly running on the input data and filtered out slowly varying changes in the signal level caused by natural body movements and breathing. Such an approach made it possible to retain signal stability and facilitate visualization of key parts of the waveform, including the P, QRS, and T waves.

As a result of using hardware filters in conjunction with software processing tools for digital signals, it became possible to detect heartbeats effectively and provide reliable measurements of the heart rate.

8.4 Data Transmission and Remote Synchronization

The reliability of the remote monitoring function in the Real-Time ECG Monitoring System was determined by analyzing the effectiveness of the communication between the sensor device, wireless module, and online monitoring system. During the testing of the system, the ECG readings and heart rate data, analyzed in the microcontroller unit, were transferred to the remote monitor via the established wireless connection. Data transmission took place continuously during the whole time of monitoring, and no disruptions occurred within the connection.

The wireless module constantly provided the server or monitoring app with the new readings and heart rate data. Moreover, each transferred package of the data had time-related parameters that allowed arranging the received data in sequence. The online monitoring system successfully received and presented the new data in the form of the corresponding chart.

The repeated testing of the data transmission demonstrated that the system could accommodate regular data updates without compromising the synchronization between the hardware component and the remote monitoring application. The acquired ECG signals and heart rate data were saved in the memory of the system or its database, thereby facilitating their retrieval at a later point in time. The performance of the communication

procedure was also observed to be stable over a period of sustained use, thus highlighting the capacity of the system to sustain long-term remote monitoring.

In summary, the outcomes of the connectivity and data transmission tests reveal that the Real-Time ECG Monitoring System provided reliable remote communication and robust data transfer capability. The effective exchange of ECG data between the local device and the remote monitoring portal indicates that the system is suitable for continuous remote monitoring of cardiac activity.

8.5 Practical Constraints and Operational Requirements

The assessment of the experimental tests revealed several crucial operational parameters that ensure proper functioning and reliability of the Real-Time ECG Monitoring System. In particular, the placement and quality of the electrodes used to measure the patient's physiological data have been identified as essential for acquiring stable and consistent results. It should be noted that using well-prepared skin surfaces and securely connected electrodes enabled stable electrical contact, allowing for a reliable acquisition of the signals during the tests. Stable connections ensured the absence of abrupt changes in the ECG graph and the possibility of measuring signals continuously.

Additionally, it has been determined that the environmental setting and arrangement of the system components significantly affected the signal acquisition performance. Experimental tests carried out in various indoor settings showed that data integrity was contingent on ensuring stable communication between the monitoring unit and the wireless network device. Specifically, when the monitoring device was positioned within the coverage area of the network, the transfer of ECG data and heart rate measurements proceeded steadily and seamlessly. In such circumstances, the system guaranteed regular updates of the acquired data on the monitoring screen without any delay.

Moreover, the system responses during the monitoring period revealed the significance of having good hardware connection and constant power supply conditions. This led to the smooth running of the system during repeated tests due to correct sensor wiring, communication links, and voltage levels. It was evident that by ensuring the right setup and conditions of the operation, signal integrity and data transmission were guaranteed for the Real-Time ECG Monitoring System.

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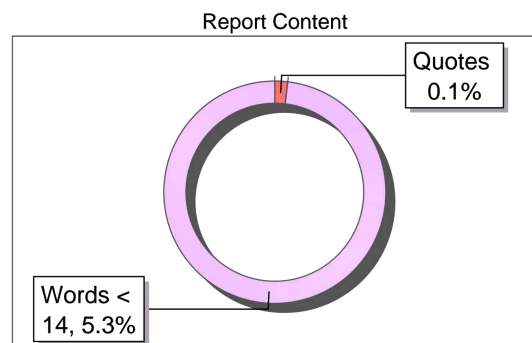
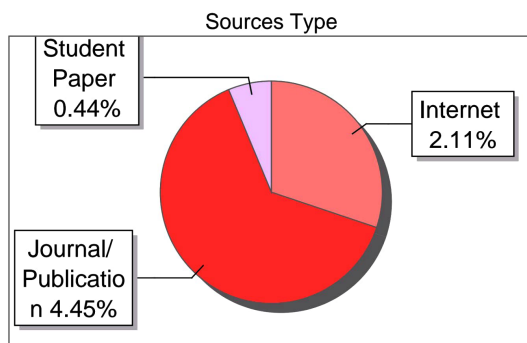
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